Estimation of Change in Population of Municipalities in PR

[ESMA 6835]: Bayesian Inference

Alejandro M. Ouslan & Tatiana Crespo

University of Puerto Rico, Mayagüez

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Contents

1 Introduction

Changes in demographics are a highly studied topic and are of great interest to governments. Understanding demographic characteristics is particularly important for regional governments, such as the municipalities in Puerto Rico. Anticipating how the population will change in the coming years is fundamental for planning infrastructure projects and public investments, especially in areas experiencing population growth, decline, or aging populations.

2 Methodology

This research uses data from the American Community Survey 5-year estimates (ACS) produced by the U.S. Census Bureau. The dataset includes population estimates from 2011 to 2023 for all 78 municipalities in Puerto Rico.

3 Resultados

3.1 Densities

3.2 Credibility Interval

3.3 Confidence Interval

3.4 Distribution

4 Predicciones

Due to the lack of statistical significance in the results regarding changes over time in municipal population, no meaningful recommendations can be made regarding the problem.

5 Conclusions

The overall results of the simple regression are inconclusive, as the variable representing year was not statistically significant. This lack of statistical significance may be due to the limitations of simple regression models. A major limitation is the violation of the assumption of independence and identical distribution (i.i.d.), since population values are highly correlated with those of the previous year. It is advisable to use a more appropriate model for time series data, such as an ARIMA model.